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PALEOMAGNETISM AND QUESTION OF ORIGINAL LOCATION OF THE PERMIAN BROOK STREET TERRANE, NEW ZEALAND

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Abstract. Over 400 rock samples from 30 sites were collected for paleomagnetic study from the volcanogenic section in the Brook Street terrane within the Takitimu Mountains in western Southland, New Zealand. The section includes igneous and sedimentary rocks of the Permian Takitimu Group and White Hill Intrusive Suite. Many of the samples show a partial or complete remagnetization in the present field because of a recent acquisition of viscous remanent magnetization. An Early Permian direction (inclination = 46.1°, declination = 257°), isolated from the Heartbreak and Chimney Peaks formations of the Takitimu Group, indicates a low paleolatitude to midpaleolatitude, position (27° ± 5°) for the Brook Street terrane. Directions from the Late Permian (?) White Hill Intrusives (inclination = 64.6°, declination = 173.3°) suggest a slightly higher paleolatitude than the Early Permian Takitimu Group directions and 70°-90° of intervening clockwise rotation. Plate reconstructions and paleomagnetic data predict a high paleolatitude for the New Zealand margin of Gondwana throughout the late Paleozoic and Mesozoic. The low paleolatitude to middle paleolatitude, implied by the Early Permian Brook Street result, together with the oceanic nature of the Brook Street arc, suggest that the Brook Street terrane is allochthonous to the margin of Gondwana. A published Late Triassic/Early Jurassic paleomagnetic pole from the adjacent Murihiku terrane indicates a high paleolatitude. This suggests that the Brook Street and Murihiku terranes are genetically distinct.

INTRODUCTION

The South Island of New Zealand comprises a broad belt of Paleozoic and Mesozoic rocks which are thought to record part of the pre-Cenozoic history of the Pacific continental margin of Gondwana. Eleven tectonostratigraphic terranes have been recognized in these rocks [Bishop et al., 1985]; the genetic relationship between these terranes is unclear. Following Coombs et al. [1976], Howell [1980] suggests that some of these terranes are exotic to the New Zealand margin of Gondwana. In contrast, MacKinnon [1983] proposed a model in which all of the terranes are autochthonous or parautochthonous to the margin of Gondwana. All of the terranes were assembled by the mid-Cretaceous [Howell, 1980].

Terranes of the Western Province (Figure 1) are mostly lower to middle Paleozoic crystalline and sedimentary rocks together with minor Mesozoic volcanic rocks and voluminous Cretaceous arc-related intrusions [Tulloch, 1983; Mattinson et al., 1986]. These are thought to represent part of the late Paleozoic and Mesozoic margins of Gondwana [Howell, 1980; MacKinnon, 1983].

The Western Province is separated from the volcanogenic marine strata of the Brook Street, Dun Mountain-Maitai, and Murihiku terranes to the east by the Median Tectonic Zone [Landis and Coombs, 1967]. These eastern terranes range from Early Permian [Waterhouse, 1964, 1982] to Late Jurassic in age [McKellar et al., 1962]. The Brook Street terrane comprises a fragmented series of Permian arc volcanics, arc-root complexes, and volcaniclastic sediments of oceanic origin [Suggate, 1978; Houghton, 1981, 1985, 1986; Sivell and Rankin, 1983; Bishop et al., 1985; Frost and Coombs, 1987]. The Murihiku terrane is a thick sequence of arc-derived Triassic and Jurassic sediments with a continental component which have been folded into a broad regional syncline [Boles, 1974]. Until recently, the Brook Street and Murihiku terranes were regarded as being in sedimentary contact. Coombs et al. [1976], Howell [1980], MacKinnon [1983], and Bishop et al. [1985] have considered the Brook Street terrane to have provenance links to the Maitai and Murihiku terranes. The Brook Street and Murihiku terranes have recently been shown to be only in fault contact with each other [Landis et al., 1987; Landis, 1987]. Further, neodymium isotope data indicates that the Brook Street terrane has a mantle source, while the

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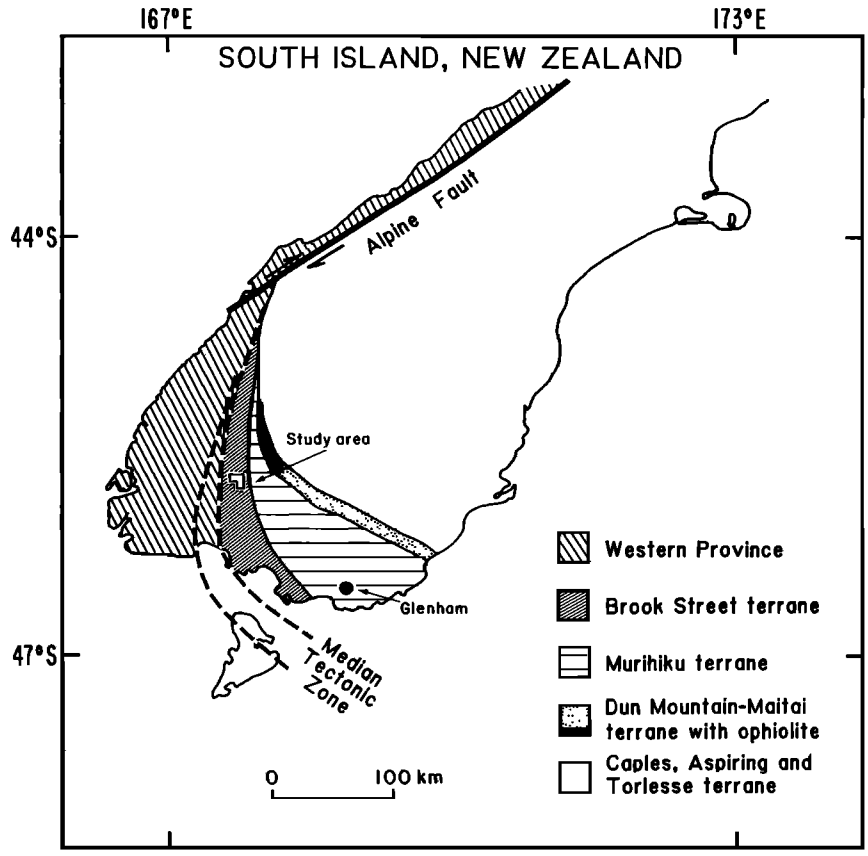


Fig. 1. Map showing terranes of the Eastern Province in the southern part of the South Island, New Zealand (modified from Bishop et al. [1985]. The locality for Figure 2 is outlined.

Murihiku terrane has a continental origin [Frost and Coombs, 1987; D. S. Coombs, personal communication, 1988]. The provenance links between the Brook Street and the Maitai terranes which have been reported on in the past are based on nonunique rocks or fossils which had a wide geographic distribution. There are no clearly demonstratable links between the Brook Street and the other terranes prior to later Cretaceous time (D. Kimbrough, personal communication, 1989; C. A. Landis and D. S. Coombs, unpublished data, 1989).

The goal of our study is to use paleomagnetic data to constrain the relationship of the Brook Street terrane with surrounding terranes and Gondwana. According to plate reconstructions constrained by paleolatitudes [Molnar et al., 1975; Weissel and Hayes, 1977] and paleomagnetic data from Gondwana fragments [Embleton, 1981; Thompson and Clark, 1982; Irving and Irving, 1982], the late Paleozoic and Mesozoic southwest Pacific margin of Gondwana occupied a high-latitude position near the south pole. Prior to the present study, the oldest paleomagnetic pole from New Zealand was from Late Triassic/Early Jurassic rocks of the Glenham Porphyry [Grindley et al., 1980] in the Murihiku terrane. The Glenham Porphyry result indicated a high paleolatitude (66°S) for the Murihiku terrane, which is consistent with results from Australia and Antarctica [Embleton, 1981; Thompson and Clark, 1982; Irving and Irving, 1982].

GEOLOGIC SETTING

The Takitimu Mountains comprise a 14-km thick sequence of east facing, homoclinal, steeply dipping, and in places overturned, volcanic, volcaniclastic, and shallow intrusive rocks of Permian age (Figure 2) [Houghton, 1981]. The age of the initiation of tilting is unclear, but it is pre-Late Jurassic [C. A. Landis, work in progress] and probably continued through the Cenozoic. The section is divided into two distinct groups: the stratified Early Permian Takitimu Group and the Permian White Hill Intrusive Suite [Houghton, 1981, 1986]. Both groups are considered part of the larger Brook Street tectonostratigraphic terrane.

The Takitimu Group comprises a thick sequence of volcanic and volcaniclastic rocks, dominated by volcaniclastic sediments, deposited in an active island arc-related basin [Houghton, 1981]. No continental detritus is found in the section, which suggests an oceanic origin. This conclusion is further supported by neodymium isotope data [Frost and Coombs, 1987] which show oceanic values. The age of the section ranges from Sakmarian near the base to Artinskian near the top [Waterhouse, 1964]. All the paleomagnetic samples cited in Table 1 are from rocks correlated by Waterhouse [1982] as later Early Permian or Baigendzinian (Artinskian) in age. The group is fault bounded against Tertiary sediments at

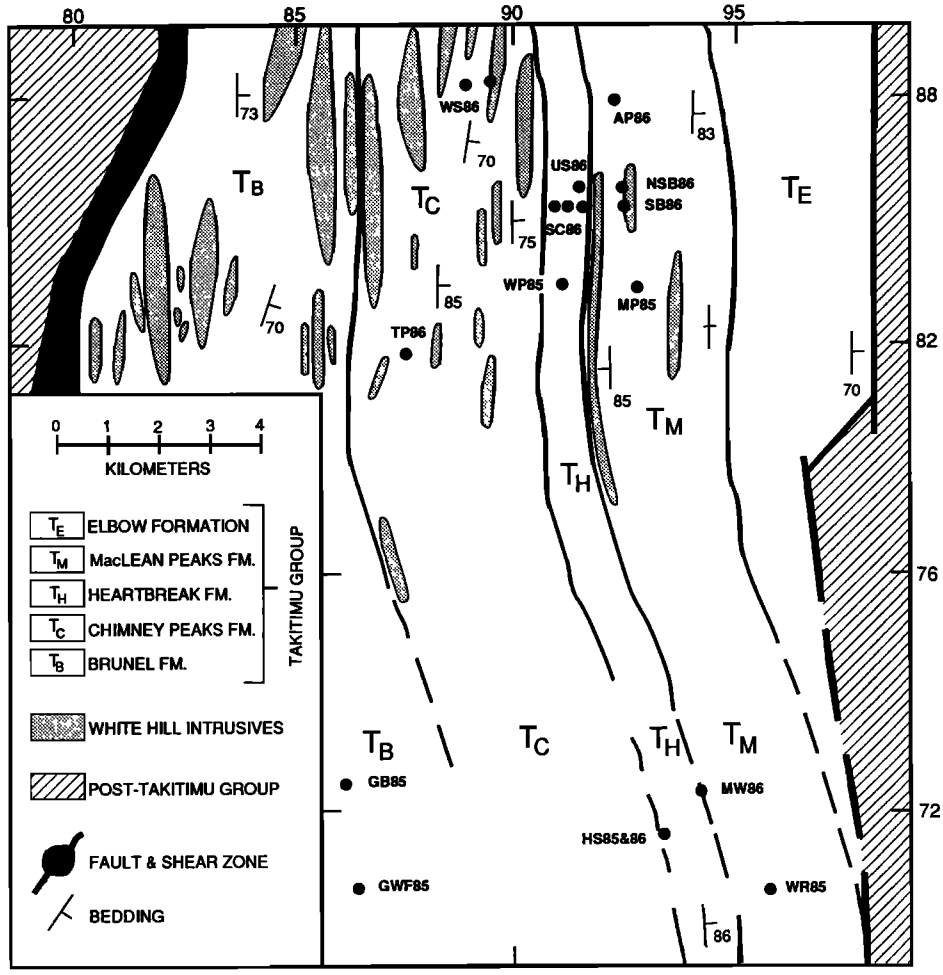


Fig. 2. Generalized geologic map of the central Takitimu Mountains and paleomagnetic site locations for this study. The northern half is from Houghton [1981]; boundaries in the southern half are reconnaissance extrapolations; the east side is from recent mapping. Coordinates are from the national thousand-yard grid of the 1:63,360 topographic series (NZMS 1).

the base [Houghton, 1981] and against the Murihiku terrane at the top [Landis et al., 1987].

A sequence of gabbroic to quartz dioritic subvolcanic intrusions, referred to as the White Hill Intrusive Suite [Houghton, 1981, 1986], intrudes the Takitimu Group. These intrusions make up about 20% of the total stratigraphic thickness in the Takitimu Mountains and are concordant or nearly concordant with bedding. They are thought to have been emplaced while the Takitimu Group was still flat lying [Houghton, 1986]. The intrusions show a similar calc-alkaline geochemical signature to the Takitimu Group and have been interpreted as originating from a similar magma source [Houghton, 1985]. The age of the intrusives is unclear. They must be Early Permian (Artinskian) or younger because they intrude the Takitimu Group rocks of that age. The White Hill Intrusives yielded a Jurassic-Triassic K/Ar age, which probably reflects the age of uplift [Houghton, 1977]. The White Hill Intrusives are cut by the MacKinnon Peak Intrusives, which yield a Triassic 231-242 Ma K/Ar age on hornblendes [Houghton, 1987]. The difference in age between the Artinskian fossils in the Takitimu Group and the K/Ar age leaves a 20-40 m.y. uncertainty in the age of the White Hill Intrusives, although Houghton [1986] feels that the White Hill Intrusive Suite was emplaced in Late Permian time.

Metamorphism is generally low grade and ranges from zeolite facies in the stratigraphically higher east to prehnite-pumpellyite or prehnite-actinolite facies in the deeper west [Houghton, 1982]. According to Liou et al. [1987], the zeolite to prehnite-actinolite facies transition occurs at about 230°C, and hence regionally developed temperatures in the sampled area are unlikely to have exceeded that value. All of the samples listed in Table 1 come from rocks within the zeolite facies. Parts of the Takitimu Group possibly experienced short-lived elevated temperatures associated with the subvolcanic plutonism, followed by a longer period of burial under a moderate geothermal gradient affecting the whole

Table 1. Site Results From The Takitimu Mountains

Site Identification	Geologic Formation	Rock Type	Grid Coordinates [a]	Number Averaged	Uncorrected Inc	Uncorrected Dec	Bedding Attitude		Corrected Inc	Corrected Dec	Corrected a95	Corrected kappa	VGP Lat	VGP Long	VGP dp	VGP dm
Takitimu Group Results:																
WS86-2	Chimney Peaks	Basalt	889888	6	-54.4	242.2	006/110	E [d,e]	43.2	251.6	19.9	14.9	-30.2	84.1	15.4	24.8
SC86-1	Heartbreak	Basalt Flow	851904	13	-44.1	250.2	010/90	E [e]	34.5	255.1	13.4	16.1	-23.8	81.1	8.8	15.4
SC86-2	Heartbreak	Basalt Flow	851905	5	-30.8	271.7	010/90	E [e]	48.2	261.7	14.1	27.1	-25.9	94.1	12.1	18.3
SC86-3 [b]	Heartbreak	Basalt Flow	851905	9	-74.6	322.6	010/90	E [e]	12.2	294.6	6.9	63.8	----	----	----	----
SC86-4 [b]	Heartbreak	Pillow Basalt	851908	7	-60.1	269.6	008/78	E [d]	-2.1	276.5	10.5	33.4	----	----	----	----
SC86-5 [c]	Heartbreak	Pillow Basalt	852908	8	-40.9	278.2	008/78	E [d]	50.1	272.5	10.7	24.2	-19.9	102.1	9.6	14.3
SC86-6 [c]	Heartbreak	Pillow Basalt	852909	8	20.1	264.6	002/78	E [f]	74.8	154.1	16.7	17.1	-68.3	202.1	27.1	30.4
HS85-1	Heartbreak	Pillow Basalt	720940	15	-35.3	260.5	350/86	E [f]	42.7	263.3	14.6	13.5	-21.9	91.4	11.1	18.1
HS86-1	Heartbreak	Pillow Basalt	719939	9	-41.6	263.9	350/86	E [f]	42.9	260.4	9.7	29.1	-24.1	89.5	7.4	12.1
WP85-B	Heartbreak	Basalt Flow	833912	10	-12.5	248.9	003/95	E [d]	44.1	221.1	8.7	48.8	-51.8	80.6	6.8	10.9
WP85-C [c]	Heartbreak	Pillow Basalt	832913	11	-32.2	282.1	003/95	E [d]	53.9	275.5	16.6	17.1	-20.5	106.1	16.3	23.2
MW86-1 [c]	Heartbreak	Pillow Basalt	732945	6	3.9	245.7	000/90	E [d]	63.1	175.8	19.5	16.3	-86.8	278.9	24.2	30.7
MP85-1 [b]	MacLean Peaks	Andesite Flow	831932	9	-70.2	44.2	350/100	E [d]	-11.6	295.7	9.8	88.8	----	----	----	----
	Mean Early Permian Result:			7	-40.6	265.3	----		46.1	257.2	9.1	39.2	-27.8	89.6	7.3	11.4
White Hill Intrusive Suite Results:																
SB86-1	---	Diabase Sill	855920	6	-16.1	239.6	356/104	E [d,e]	64.4	183.9	19.1	13.3	-87.2	90.2	24.4	30.4
WS86-1	---	Gabbro	885895	8	2.5	234.4	006/110	E [e]	55.1	172	17.4	20.3	-78.3	314.1	17.6	24.7
	Mean Late(?) Permian Result:			4	2.7	245.8	----		64.6	173.3	10.6	75.9	-85.3	245.2	13.7	17.1

Inc, inclination, in degrees; Dec, declination, in degrees; a95, radius of cone of 95% confidence about direction, in degrees; Lat, latitude, in degrees; Long, longitude, in degrees east; dp, delta colatitude, in degrees; dm, delta meridian, in degrees.

[a] Coordinates are from the national thousand-yard grid of the 1:63360 topo series (NZMS 1).
[b] Remagnetized in the present field direction.
[c] Remagnetized in the White Hill Intrusive Suite direction.
[d] Structural correction taken from adjacent sediments.
[e] Regional structural correction used.
[f] Structural correction taken from flow top.

group [Houghton, 1982]. On outcrop scale, permeability and fractures played a major role in the degree of metamorphism [Houghton, 1982].

EXPERIMENTAL PROCEDURE

Over 400 samples from 30 sites were collected from the Takitimu Group and from the White Hill Intrusive Suite (Figure 2). In this study a site is defined as a set of samples taken from a single cooling unit or sedimentary horizon. Structural attitudes were taken from interbedded sedimentary units when possible. In areas where no structural attitude was available a regional attitude taken from maps of Houghton [1981] was used. Each sample was individually stepwise demagnetized using either stepwise alternating field (AF), stepwise thermal, or continuous thermal demagnetization [Dunn and Fuller, 1984]. Stable directions were selected using equal-area and Zijderveld [1967] vector end point plots coupled with the statistical methods of Kirschvink [1980]. Site averages were then computed using Fisher [1953] statistics.

PALEOMAGNETIC RESULTS

Many of the rocks analyzed show a natural remanent magnetization (NRM) direction which is close to or coincident with the present field direction in in situ coordinates. However, with demagnetization, stable directions were obtained from many of the samples (Figure 3a-3f). In some sites the magnetization vector varied significantly on the outcrop scale (2-10 m). Houghton [1982] found that fractures and permeability played a major role in the degree of low-grade metamorphism here. Thus variation in magnetization may also be due to the affects of fluids following fractures in the rocks. Because of this, each sample was evaluated individually for stability and direction. Samples in which there was no stable direction or where the direction was close to the present field position in in situ coordinates were rejected from the site average. Sites with an alpha 95 greater than 20° were rejected as unreliable.

Of the 30 sites sampled, 15 yield a stable direction (Table 1). Three of these show directions coincident with the present field in in situ coordinates and are rejected from tectonic analysis. The remaining 12 sites cluster into two groups (Figure 4). Ten sites are from the Takitimu Group. Eight of these show directions with similar westerly declinations, while two others (SC86-6 and MW86-1) have concordant southerly directions. All of the stable results from the Takitimu Group come from rocks Artinskian in age. Thus the two directions found in the Takitimu Group cannot be explained by a large age difference in the rocks. Two stable sites from the White Hill Intrusive Suite display southerly declinations, deflected about 90° from the Takitimu Group results (Figure 4, Table 1). Interestingly, the two anomalous site directions from the Takitimu Group (SC86-6 and MW86-1) are similar to directions of the White Hill Intrusive Suite sites (Figure 4, Table 1). These Takitimu sites were collected from highly fractured pillow lavas within 100 m of large 20- to 40 m-thick sills of the White Hill Intrusives.

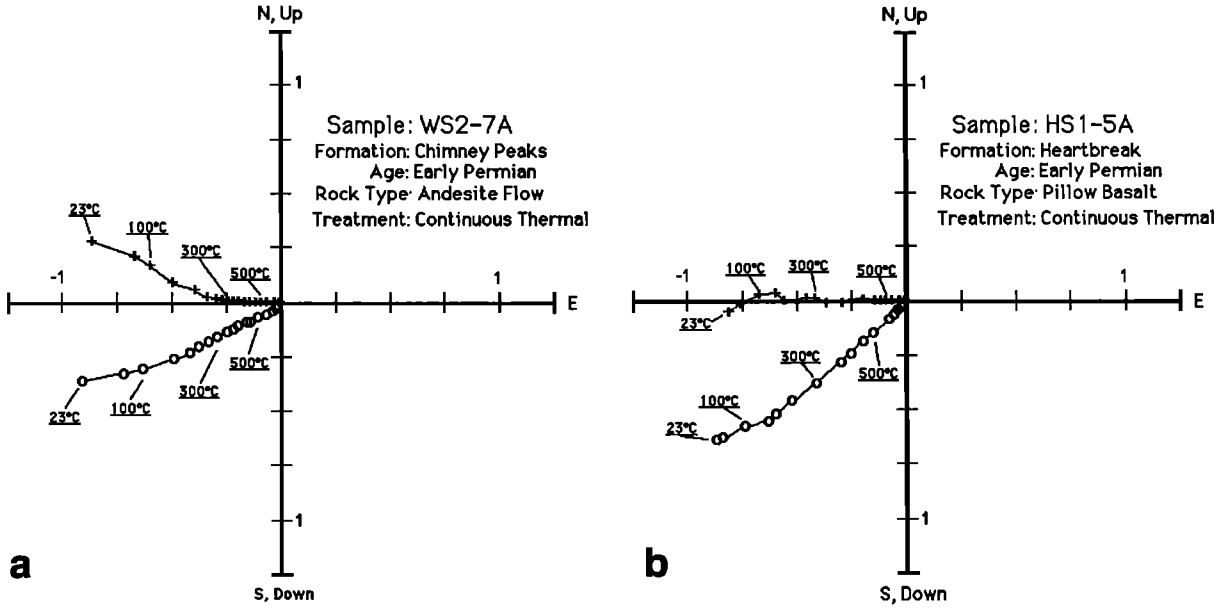


Fig. 3. Zijderveld [1967] vector end point plots for samples from the Takitimu Mountains. Crosses are the vector projection onto the declination plane; circles are the vector projection onto the inclination- north plane. (a) Continuous thermal demagnetization of sample WS2-7A from the Early Permian Chimney Peaks Formation. (b) Continuous thermal demagnetization of sample HS1-5A from the Early Permian Heartbreak Formation. (c) Continuous thermal demagnetization of sample SC2-1A from the Early Permian Heartbreak Formation. (d) Stepwise thermal demagnetization of sample SB1-5 from the Late(?) Permian White Hill Intrusive Suite. (e) Stepwise thermal demagnetization of sample WS1-5A from the Late(?) Permian White Hill Intrusive Suite. (f) Continuous thermal demagnetization of sample MW1-1A from the Early Permian Heartbreak Formation. Site is remagnetized in the White Hill Intrusive Suite direction.

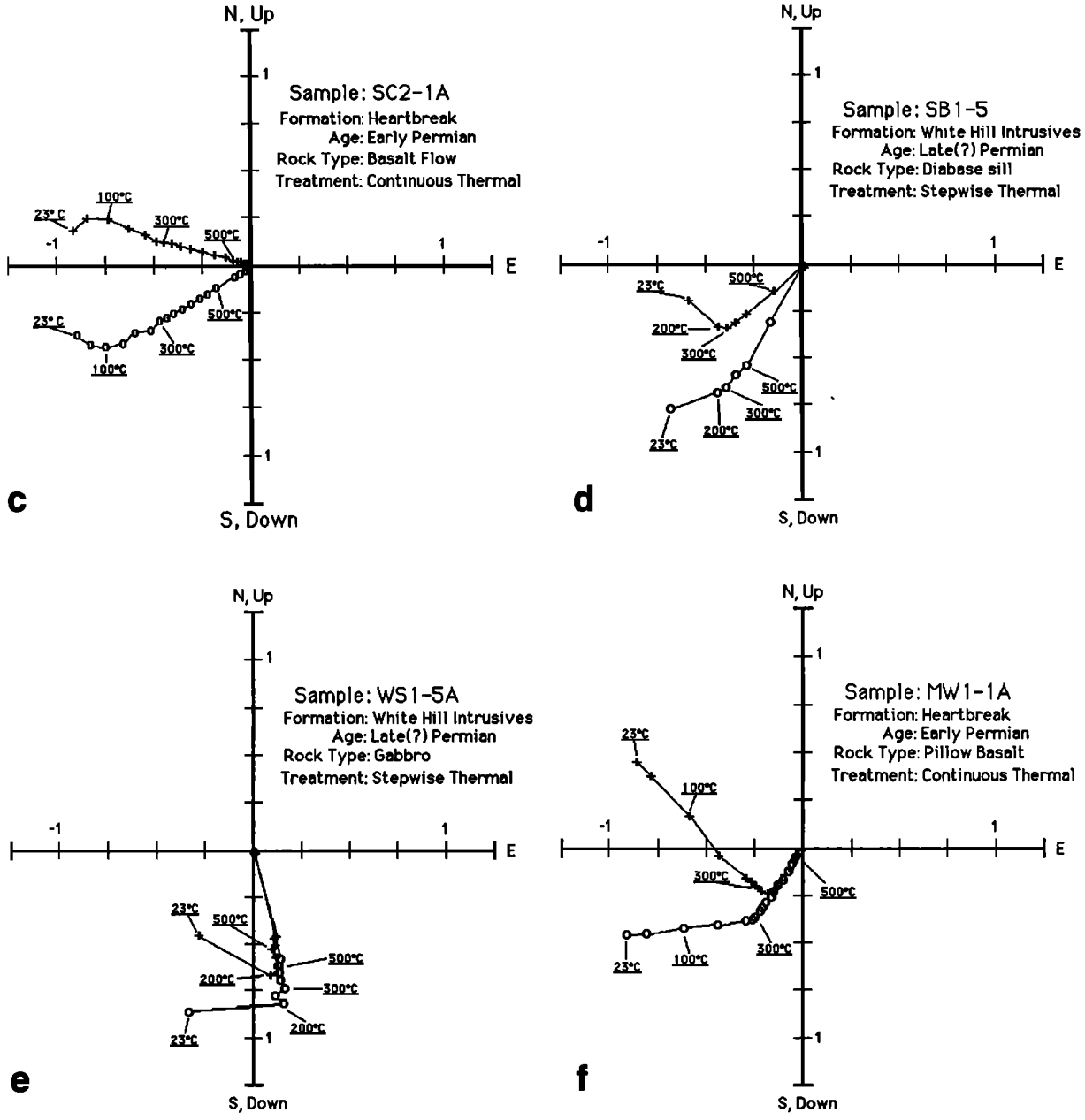


Fig. 3 (continued)

ROCK MAGNETIC EXPERIMENTS

Isothermal remanent magnetization (IRM) acquisition, IRM demagnetization, AF and thermal demagnetization of NRM, and Curie temperature experiments were run on selected samples in order to identify the magnetic carriers, their domain state, and their grain size distribution. The ultimate goal of these experiments is to try to independently ascertain whether remagnetization has occurred by identifying the type of magnetization a rock is carrying (i.e., thermal, chemical, or viscous remanent magnetization (TRM, CRM, or VRM)).

Virtually all selected samples display multidomain behavior with remanent coercivities (H_{rc}) less than 30 mT and saturation remanence achieved at 60-80 mT. This, coupled with Curie point determinations, suggests that multidomain magnetite is the magnetic carrier in samples from both the Takitimu Group and the White Hill Intrusives.

No marked differences are seen in the behavior between the samples which are magnetized in the present field direction and sites which are not, except in the J/J_0 versus demagnetization temperature curves. Samples which are in the present field direction consistently show a more widely distributed blocking

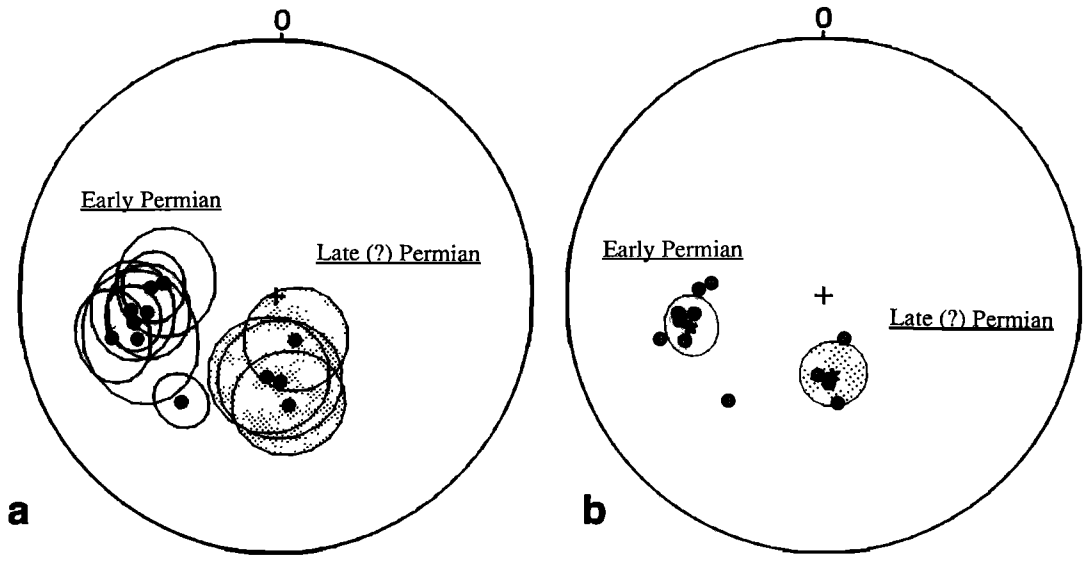


Fig. 4. (a) Lower hemisphere, equal-area stereographic projection of site mean directions and associated 95% confidence intervals. Sites with gray pattern are the Early Permian results; sites with stippled pattern are the interpreted Late Permian results. (b) Lower hemisphere, equal-area stereographic projection of site mean directions (solid circles) and the average of the Early Permian directions with associated 95% confidence intervals (star with gray pattern), and the average of the Late(?) Permian directions with associated 95% confidence intervals (star with stippled pattern).

temperature spectrum (Figure 5), suggesting that the remanence is carried by multidomain grains.

Fuller et al. [1988] suggest that plotting NRM versus IRM saturation (IRMS) through AF demagnetization may distinguish primary and secondary magnetizations, independent of direction. That is, different episodes of magnetization may be distinguishable by how efficiently the rocks are magnetized. They suggest that lavas which carry a TRM will have a NRM:IRMS ratio around 10^{-2}, while many remagnetized lavas tend to have much lower ratios around 10^{-3}. Samples displaying directions near the present field position have NRM:IRMS ratios around 10^{-3} throughout AF demagnetization. In contrast, stable samples which do not have a present field direction have NRM:IRMS ratios around 10^{-2} (Figure 6). This suggests that these rocks were magnetized by an inefficient mechanism. VRM and not chemical remanent magnetization is the probable mechanism for remagnetization in the present field because of the lack of recent alteration and because the magnetization is carried predominantly by multidomain grains.

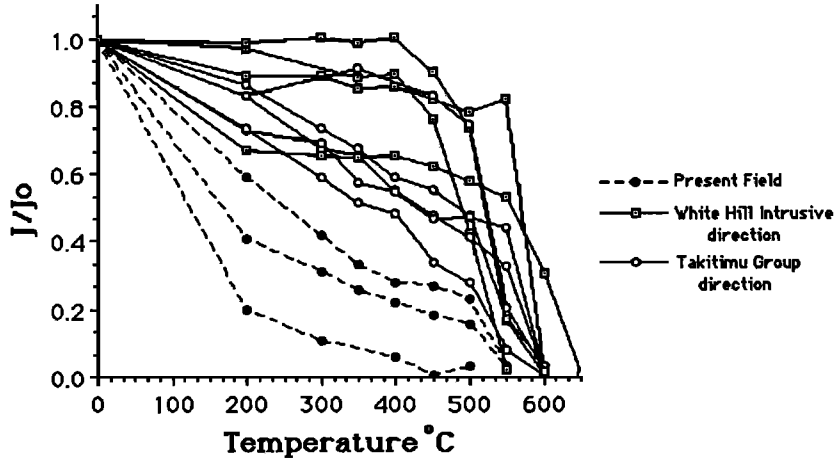


Fig. 5. J/J_0 versus demagnetization temperature for representative samples from the Takitimu mountains. Solid circles are samples remagnetized in the present field; open circles are samples in the Early Permian Takitimu Group direction; squares are samples in the White Hill Intrusive Suite direction.

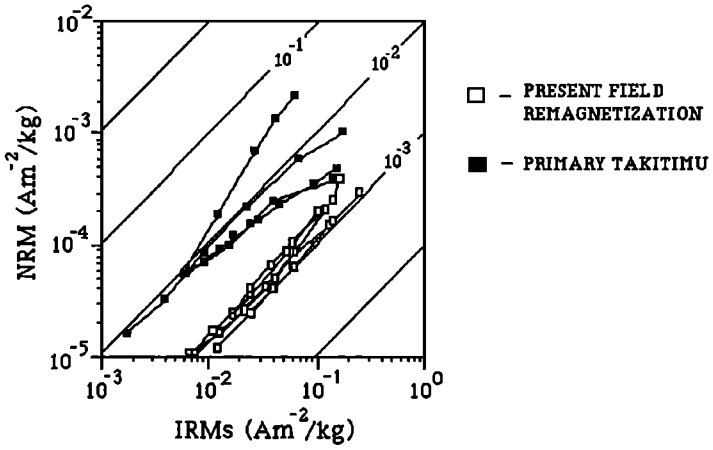


Fig. 6. NRM/IRMs plot [Fuller et al., 1988] for representative samples from the Takitimu Mountains. Solid diamonds refer to samples which display a primary direction; open squares are samples which show a present field direction.

The White Hill direction sites (SC86-6 and MW86-1) from the Takitimu Group do not show depressed NRM:IRMS ratios. Further, J/J_0 versus demagnetization temperature curves (Figure 5) show that their remanence is probably carried by fine-grained magnetite. This suggests that these rocks were remagnetized by an efficient mechanism. These rocks were probably not remagnetized by partial TRM (PTRM) because the small size of the intruding sills makes it unlikely that they would have raised the temperature significantly. Further, the metamorphic mineral assemblages in the Takitimu Group indicate that temperatures probably did not exceed 200° or 300°C [Houghton, 1982]. Because the rocks from sites MW86-1 and SC86-6 are highly fractured, they may have been remagnetized by precipitation of fine magnetite during moderately elevated temperatures and fluid migration, caused by the emplacement of the White Hill Intrusives.

DISCUSSION

Certainly reliable paleomagnetic data are difficult to obtain in active orogenic belts such as New Zealand. The tectonic complexity of the region here is reflected in magnetization of the rocks, making it difficult to confidently isolate stable directions. A strong present field remagnetization has either partially or completely reset the magnetization in many of the rocks in the Takitimu Mountains. The mechanism for this was probably the acquisition of VRM and not chemical or mineralogic changes in the rocks. This is supported by (1) the low blocking temperatures for the present field remagnetized samples which suggests that multidomain grains are carrying the remanence, (2) thin section examinations which show no significant differences in alteration between present field remagnetized and primary magnetized samples, (3) the presence of magnetite as the magnetic carrier in both the present field remagnetized and primary rocks, and (4) a depressed NRM:IRMS ratio of around 10^{-3} for the present field remagnetized samples.

The eight sites from the Takitimu Group with concordant westerly declinations yield a mean tilt-corrected direction of inclination = 46.1°, declination = 257.0° (alpha 95 = 9.1°, kappa = 39.2). There is no significant difference in the alpha 95s and kappas between the in situ and the tilt-corrected directions because the beds are roughly homoclinally dipping. Structural attitudes for four of these sites came from adjacent sedimentary units, two were regional attitudes from Houghton [1981], and only two were from flow tops (Table 1). Thus any initial dip of the flows is probably not biasing the data. The large within site scatter in each site (Table 1) suggests that the White Hill and/or present field remagnetizations are pervasive. The relatively high kappa suggests that secular variation is not completely averaged. We interpret the Takitimu direction to be an Early Permian age for the following reasons: (1) Single components of magnetization are isolated at higher demagnetization levels in many of the rocks (Figures 3a-3c), (2) all of the rocks are magnetized in a reversed polarity; this is consistent with the age of the rocks which lie in the Permo-Carboniferous long reversed period [Harland et al., 1982], (3) the NRM:IRMS ratios of samples which show this direction are around 10^{-2}, suggesting that these rocks carry a TRM or a CRM, and (4) if the rocks were magnetized posttilting, then the age of the magnetization is post-Late Jurassic and the in situ magnetic direction (inclination = -37.4°, declination = 262.7°) implies a paleolatitude around 20°S. It is difficult to accept a 20°S position in view of the fact that all terranes were accreted by the mid-Cretaceous [Bishop et al., 1985] and also because global reconstructions [e.g., Molnar et al., 1975] show New Zealand occupying a position around 65°S in the Cretaceous and moving northward through the Tertiary. Therefore the magnetization probably predates the tilting.

The two stable site results from the White Hill Intrusive Suite suggest that these rocks were magnetized at a different time from the Takitimu Group. These results further infer that the Takitimu Group rocks experienced substantial clockwise rotation prior to the intrusion event. The White Hill Intrusive magnetization (inclination = 64.6°, declination = 173.3°) probably predates the tilting for the same reasons as above, because the in situ direction (inclination = 2.7°, declination = 245.8°) would imply a paleolatitude near the equator. The age of magnetization is probably Late Permian because the White Hill Intrusive Suite is crosscut by the MacKinnon Peak Intrusives with a Triassic K-Ar age, and is also reversely magnetized. The two sites from the Takitimu Group which have White Hill Intrusives directions were probably remagnetized by the emplacement of the White Hill Intrusives.

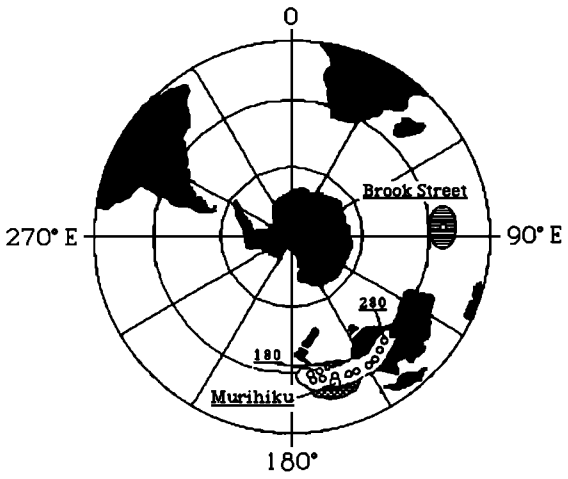


Fig. 7. Late Paleozoic and Mesozoic synthetic APW path for Gondwana (data from Thompson and Clark [1982]) transferred from Africa to New Zealand using rotation poles from Lawver and Scotese [1987]. Individual points represent 10 m.y. increments. The Murihiku results are from the Glenham Porphyry [Grindley et al., 1980].

However, it is not clear what the mechanism for this remagnetization may have been.

POLAR WANDER PATHS

An Early Permian pole (latitude = 27.8°S, longitude = 089.6°E, delta colatitude = 7.3, delta meridian = 11.4) for the Brook Street terrane is obtained from the Takitimu Group paleomagnetic results. The oldest previously published pole from New Zealand is a Late Triassic/Early Jurassic result from the Murihiku terrane [Grindley et al., 1980]. The lack of older poles and a New Zealand apparent polar wander (APW) path make it impossible to test whether the Brook Street terrane is allochthonous. However, an APW path for Gondwana from Thompson and Clark [1982] was transferred (using rotation poles from Lawver and Scotese [1987]) to New Zealand (Figure 7). The Permian-Jurassic rocks of the South Island have been oroclinally bent through approximately 90°. This is particularly evident in the Dun Mountain-Maitai Terrane (Fig. 1). Some question remains regarding the timing of bending. Oliver et al. [1979], assuming that the orocline involves Mesozoic greywackes in mid-Canterbury, conclude from paleomagnetic data that the bend formed mainly during Early Cretaceous time. In contrast, Norris (1979) shows from finite strain analysis that the oroclinal fold only affected a strip of rocks adjacent to the Alpine Fault and that bending east of the Alpine Fault is compatible with known late Cenozoic displacements. Regardless of age, the oroclinal fold would displace the apparent pole position for the Takitimu Group. However, rotations about a vertical axis will not bring the Brook Street pole into agreement with the synthetic New Zealand APW path, implying that latitudinal motion has occurred between the Brook Street terrane and the New Zealand margin of Gondwana. The predicted paleolatitude from the Brook Street terrane (27° ± 5°) is compared with the observed paleolatitude for the New Zealand margin of Gondwana in Figure 8. Clearly, the observed Early Permian paleolatitude for the Brook Street terrane is much lower than the predicted Gondwana paleolatitude, suggesting that this terrane was either allochthonous to the margin of Gondwana or came from further north but near to the Gondwana margin. In contrast to this, the Late Triassic/Early Jurassic pole from the Murihiku terrane [Grindley et al., 1980] is close to the Gondwana APW path (Figures 7 and 8), suggesting that this terrane was deposited near the margin of Gondwana. These data suggest that the Brook Street and Murihiku terranes are two distinct tectonostratigraphic terranes; this is strongly supported by field mapping evidence presented by Landis et al. [1987].

PALEOBIOLOGICAL CONSTRAINTS

A clear New Zealand link with Gondwana is the plant fossil *Glossopteris*, which has been found as transported leaves in mollusk- and brachiopod-bearing limestone of the Productus Creek Group [Mildenhall, 1970] in a fault sliver of the Brook Street terrane between the Takitimu Group and Murihiku terrane. New Zealand Permian faunas have been studied extensively by Waterhouse [1964, 1982], who has mapped brachiopod zones within the Takitimu Mountains and has used them for international correlation and for conclusions regarding Early Permian paleoclimate and paleolatitude. Waterhouse has argued for strong climatic fluctuations during deposition of the Takitimu Group, and he interpreted these as representing glacial and interglacial episodes. He suggested that the rocks originated within an island arc geosyncline flanking the Antarctic segment of Gondwana. Latitude was estimated at 50-55°S [Waterhouse, 1982]. Although Waterhouse stressed evidence for wide climatic variations, faunas from the interval sampled for our paleomagnetic study were interpreted as living under warm conditions. This conclusion is compatible with the paleolatitude inferred from the paleomagnetic data. However, the paleomagnetic data do not explain the cool water environments inferred for earlier and later Permian faunas. Retallack [1987] has indicated a high paleolatitude origin for flora in the Murihiku terrane, a conclusion compatible with the 66°S Late Triassic-Early Jurassic Murihiku paleolatitude inferred by Grindley et al. [1980].

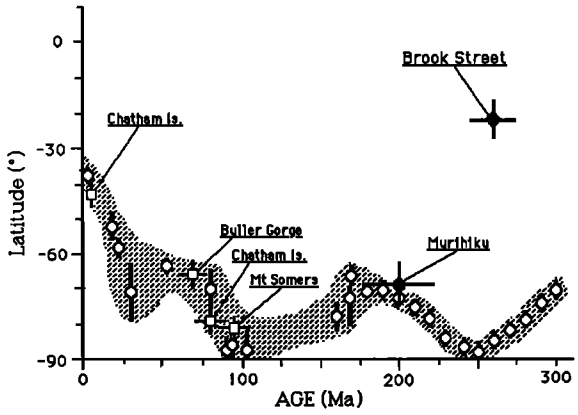


Fig. 8. Predicted latitude versus. time plot for the New Zealand margin of Gondwana. Data from Embelton [1981], Thompson and Clark [1982], and Irving and Irving [1982]. Chatham Island data from Grindley et al. [1977]; Buller Gorge data from Grindley and Oliver [1979]; Mount Somers data from Oliver et al. [1979].

CONCLUSIONS

Despite the magnetic complexity of the rocks studied, we feel that we have isolated an Early Permian direction for the Brook Street terrane. This direction suggests that this terrane may have an allochthonous origin to the New Zealand margin of Gondwana. This is counter to many previously published models [Coombs et al., 1976; MacKinnon, 1983]. The paleomagnetic data on their own are not strong enough to support such an important conclusion, which may affect the interpretation of much of the geology in New Zealand. However, recent work done in New Zealand also suggests that there is no provenance linking between the Brook Street and any of the other adjacent terranes [Landis 1987; Landis et al., 1987; Frost and Coombs, 1987] . More paleomagnetic data are needed from the various terranes in New Zealand, but we hope that these data will stimulate further discussion and testing of ideas in order to understand the complex nature of an orogenic belt like New Zealand.

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